

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester IV)**  
**EnterPrise Java**  
**Practical-IX**

|                               |                                    |
|-------------------------------|------------------------------------|
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| Class:TYIT                    | Batch:01                           |
| Date of Assignment:12-09-2026 | Date/Time of Submission:12-09-2026 |

**9. Implement the following JPA applications.**

**a. Develop a simple Inventory Application Using JPA.**

**Code:-CheckInventory.jsp**

```
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.persistence.Persistence"%>
<%@page import="javax.persistence.EntityManager"%>
<%@page import="javax.persistence.EntityManagerFactory"%>
<%@page import="javax.persistence.EntityTransaction"%>
<%@page import="myAPP.Inventory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%
```

```
EntityManagerFactory eMF =
```

```
    Persistence.createEntityManagerFactory("INVJPAPU");
```

```
EntityManager eM = eMF.createEntityManager();
```

```
EntityTransaction eT = null;
```

```
List<Inventory> inventoryList;
```

```
String submit = request.getParameter("AddToInventory");
```

```
if (submit != null && submit.equals("AddToInventory")) {
```

```
    try {
```

```
        String itemname =
```

```
            request.getParameter("itemname");
```

```
        String description =
```

```
            request.getParameter("description");
```

```
        String quantity1 =
```

```
            request.getParameter("quantity");
```

```
        int quantity =
```

```
            Integer.parseInt(quantity1);
```

```
        String addedDate =
```

```
            new java.util.Date().toString();
```

```
        Inventory inv = new Inventory();
```

```
        inv.setItemName(itemname);
```

```
        inv.setDescription(description);
```

```
        inv.setQuantity(quantity);
```

```
        inv.setAddedDate(addedDate);
```

```
        eT = eM.getTransaction();
```

```
        eT.begin();
```

```
        eM.persist(inv);
```

```
        eT.commit();
```

```
    } catch (RuntimeException e) {
```

```
        if (eT != null && eT.isActive()) {
```

```
            eT.rollback();
```

```
        throw e;}}
```

```
inventoryList =
```

```
    eM.createQuery("SELECT i FROM Inventory i",
```

```
        Inventory.class)
```

```
        .getResultList();
```

```
eM.close();
```

```

%>
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Inventory</title>
</head>
<body>
<table border="1">
<tr>
  <td colspan="5" align="center">
    <b>
      Click
      <a href="index.jsp">here</a>
      to add items in the Inventory.
    </b>
  </td>
</tr>
<tr>
  <th>Item ID</th>
  <th>Item Name</th>
  <th>Description</th>
  <th>Quantity</th>
  <th>Added Date & Time</th>
</tr>
<%
Iterator<Inventory> iterator =
    inventoryList.iterator();
while (iterator.hasNext()) {
    Inventory obj =
        iterator.next();
%>
<tr>
  <td><%=obj.getItemID()%></td>
  <td><%=obj.getItemName()%></td>
  <td><%=obj.getDescription()%></td>
  <td><%=obj.getQuantity()%></td>
  <td><%=obj.getAddedDate()%></td>
</tr>
<%}
%>
</table>
</body>
</html>

```

### Code:-Index.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr>
  <td colspan="2" align="center">
    <b>Add the item in Inventory</b>

```

```

        </td>
    </tr>
    <tr>
        <td>Item Name:</td>
        <td>
            <input name="itemname" maxlength="25" size="50">
        </td>
    </tr>
    <tr>
        <td>Description:</td>
        <td>
            <textarea rows="5" cols="36"
                name="description"></textarea>
        </td>
    </tr>
    <tr>
        <td>Quantity:</td>
        <td>
            <input name="quantity" maxlength="4" size="5">
        </td>
    </tr>
    <tr>
        <td colspan="2">
            <input type="submit"
                name="AddToInventory"
                value="AddToInventory">
        </td>
    </tr>
</table>
</form>
</body>
</html>

```

### Code:-Inventory.jsp

```

package myAPP;
import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
@Entity
@Table(name = "inventory")
public class Inventory {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "itemid", unique = true, updatable = false)
    private Integer itemid;
    @Column(name = "itemName")
    private String itemName;
    @Column(name = "description")
    private String description;
    @Column(name = "quantity")
    private Integer quantity;
    @Column(name = "addedDate")
    private String addedDate;
    public Inventory() {}
    public Integer getItemID() {
        return itemid;}
}

```

```

public void setItemID(Integer itemid) {
    this.itemid = itemid;}
public String getItemName() {
    return itemName;}
public void setItemName(String itemName) {
    this.itemName = itemName;}
public String getDescription() {
    return description;}
public void setDescription(String description) {
    this.description = description;}
public Integer getQuantity() {
    return quantity;}
public void setQuantity(Integer quantity) {
    this.quantity = quantity;}
}
public String getAddedDate() {
    return addedDate;}
}
public void setAddedDate(String addedDate) {
    this.addedDate = addedDate;}
}
}

```



